

Understanding the Equity Summary Score Methodology

The Equity Summary Score provides a consolidated view of the ratings from a number of independent research providers on Fidelity.com. Some stocks are not rated by all research providers. Since the model uses a number of ratings to arrive at an Equity Summary Score, only stocks that have four or more firms rating them have an Equity Summary Score. It uses the providers' relative, historical, recommendation performance along with other factors to give you an aggregate, historical accuracy-weighted indication of the independent research firms' stock sentiment.

As discussed in detail below, this single stock score and associated sentiment is provided by LSEG StarMine focused primarily on building quantitative factor models for institutional investors. It is calculated by normalizing third-party research providers' ratings distributions (making them more comparable) and weighting each provider's rating in the final score based on historical accuracy. Equity Summary Scores for the 1,500 largest stocks by market capitalization are force ranked to help ensure a consistent ratings distribution. This means that there will be a diversity of scores provided by the model, thereby assisting investors in evaluating the largest stocks (in terms of capitalization), which typically make up the majority of Fidelity's investors' portfolios. Finally, smaller cap stocks are then slotted into this distribution without a force ranking, and may not exhibit the same balanced distribution. LSEG StarMine updates Equity Summary Scores daily based on the ratings provided to it by the independent research providers after the close of each trading day.

How are Equity Summary Scores calculated?

The **LSEG StarMine** model takes the multiple standardized ratings of the research providers and creates a single **Equity Summary Score/Sentiment** using the following steps:

- 1. Normalize** – Look at the research providers' buy and sell ratings distributions to understand which ratings are scarce and therefore more important.
The distribution of ratings from each of the independent research firms are normalized to make them more comparable with each other. For example, some research providers may issue a large number of buy recommendations and few sell recommendations, or vice versa. LSEG StarMine adjusts for this by overweighting "scarce" ratings and underweighting "plentiful" ratings. By normalizing the distribution of ratings, the model can recognize the "scarcity value" of ratings that are infrequently given which adds additional information to the model.
- 2. Weight** – Look at the 24 month relative firm/sector ratings accuracy and use that information to determine which firms' ratings have the most weight in the aggregated Equity Summary Score.
Since 2009 on Fidelity.com, LSEG StarMine has run its sophisticated scoring system to facilitate a fair comparison of research provider recommendation performance across widely disparate industries and market conditions. The StarMine Relative Accuracy Score for each research provider uses the past performance of the provider's individual stock recommendations with that of its peers in each sector to calculate a statistical aggregation ranging from 1 to 100. It is calculated over a 24-month period based on the performance of a research firm within a given sector against its peer set of other firms in the market rating stocks in this sector. The calculation is analogous to a "batting average score", that is how often stocks rated "buy" outperform the market and stocks rated "sell" underperform the market as a whole. To get a score higher than 50, the industry-relative return of a firm's recommendations within a sector must, when taken together, be greater than those of the median provider. The StarMine Relative Accuracy Score is used in the Equity Summary Score model to help weight the individual provider stock recommendations in the aggregated Equity Summary Score.
- 3. Calculate** – The normalized analysts' recommendations and the accuracy weightings are combined to create a single score. For the largest 1,500 stocks by market capitalization, these scores are then forcibly ranked against all the other scores to create a standardized Equity Summary Score on a scale of 0.1 to 10.0 for the 1,500 stocks. This means that there will be a uniform distribution of scores provided by the model thereby assisting investors in evaluating the largest stocks (in terms of

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capitalization), which typically make up the majority of individual investors' portfolios. Finally, smaller cap stocks are then slotted into this distribution without a force ranking, and may not exhibit the same balanced distribution.

The Equity Summary Score and associated sentiment ratings by LSEG StarMine are:

- 0.1 to 1.0 - very bearish
- 1.1 to 3.0 - bearish
- 3.1 to 7.0 - neutral
- 7.1 to 9.0 - bullish
- 9.1 to 10.0 - very bullish

Other Important Model Factors:

- An Equity Summary Score is only provided for stocks with ratings from four or more independent research providers.
- New research providers are ramped in slowly by LSEG StarMine to avoid rapid fluctuations in Equity Summary Scores. Independent research providers that are removed from Fidelity.com will similarly be ramped out slowly to avoid rapid fluctuations.

Notes on Using the Equity Summary Score:

The Equity Summary Score and sentiment ratings are ratings of relative, not absolute forecasted performance. The LSEG StarMine model anticipates that the highest rated stocks, those labeled "Very Bullish" **as a group**, may outperform lower rated groups of stocks. In a rising market, most stocks may experience price increases, and in a declining market, most stocks may experience price declines.

- Proper diversification within a portfolio is critical to the effective use of the Equity Summary Score. Individual company performance is subject to a broad range of factors that cannot be adequately captured in any rating system.
- Larger differences in Equity Summary Scores may lead to differences in future performance. The sentiment rating labels should only be used for quick categorization. An 8.9 *Bullish* is closer to a 9.1 *Very Bullish* than a 7.1 *Bullish*.
- For a customer holding a stock with a lower Equity Summary Score, there are many important considerations (for example, taxes) that may be much more important than the Score.
- The Equity Summary Score does not predict future performance of underlying stocks. The Equity Summary Score model has only been in production since August 2009 and therefore no assumptions should be made about how the model will perform in differing market conditions.

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Important Information on Monthly Performance Calculations by LSEG StarMine

- The set of covered stocks and ratings are established as of the second to last trading day of a given month. For a stock to be included in the scorecard calculations, it must have an Equity Summary Score as of the second to the last trading day of the month. The positions are assumed to be entered into on the last trading day of the month, and, if necessary, exited on the last trading day of the next month.
- The Scorecard calculations use the closing price as of the last trading day of the month. The Scorecard calculations assume LSEG StarMine exits old positions and enters new ones at the same time at closing prices on the last trading day of a given month. The calculations assume 100% investment at all times.
- The 1-Year total return by Market Cap table breakpoints for the largest 500 stocks (large cap), the next 400 (mid cap), and the next 600 (small cap), are also established as of the end of trading on the second to the last trading day of a given month.
- The calculation of performance assumes an equal dollar weighted portfolio of stocks, i.e. the theoretical investment allocated to each stock is the same.
- Performance in a given month for a given stock is calculated as [starting price (starting price meaning closing price as of the last day of trading of the prior month) less the ending price, divided by the starting price.] Prices incorporate any necessary adjustments for dividends and corporate actions (e.g. splits or spinoffs).
- The performance of a given tier of rated stocks is calculated by adding up the performance of all stocks within that given tier, then dividing by the total number of stocks in a given tier.
- The process for the next month begins again by looking at Equity Summary Scores as of the second-to-last trading day of the new month, placing stocks into their given tiers, and starting the process all over again.
- It is important to note that the “theoretical” portfolio rebalancing process that LSEG StarMine performs between the end of one month and the beginning of the next month is, for the purposes of the scorecard, a cost-free process. **This means that no commissions or other transaction costs (e.g. bid/ask spreads) are included in the calculations.**
- If a customer attempted to track portfolios of stocks similar to those included in the scorecard, their returns would likely differ due to transaction costs as well as different purchase and sale prices received when buying or selling stocks.

About LSEG StarMine

LSEG StarMine is focused primarily on building quantitative factor models for institutional investors. LSEG StarMine's equity analytics and research management tools help investment firms around the globe generate alpha and process equity information more efficiently. They are one of the largest and most trusted sources of objective equity research performance scores. Their performance scoring helps investors anticipate trends in analyst sentiment, predict surprises, evaluate financial statements for measures of earnings quality, and more.

Using the Equity Summary Score

There are many ways to use the Equity Summary Score. You can use it as a screening criterion, to help identify stocks you may want to include or exclude from further analysis, in conjunction with other criteria. You can also use it to monitor the consolidated opinion of the independent research providers that are following the stocks currently in your portfolio.

The Equity Summary Score from LSEG StarMine is not:

- A Fidelity rating. As with the other content provided in the stock research section of Fidelity.com, the Equity Summary Score comes from an independent third-party, LSEG StarMine.
- Simply an average analyst rating. The Equity Summary Score is the output of a model whose inputs are the ratings of the independent research providers (IRPs).
- A buy or sell rating. It is a calculated expression of the overall "sentiment" of the IRPs who have provided a rating on a stock.
- Directly comparable to a consensus rating. A consensus rating is generally a simple "average" rating, while the Equity Summary Score is a model-calculated value. First Call Consensus Recommendation is provided where available along with the Equity Summary Score for a stock. First Call Consensus Recommendation is provided by Thomson Reuters, an independent third-party, using information gathered from contributors. The number of contributors for each security where there is a consensus recommendation is provided. Each contributor determines how their individual recommendation scale maps to the standardized Thomson Reuters First Call scale of 1-5.

Who are the Independent Research Providers and how does LSEG StarMine receive their ratings?

Fidelity's brokerage customers enjoy one of the broadest sets of independent research providers (IRPs) available for evaluating stocks. The Equity Summary Score provides a consolidated view of the ratings from a number of independent research providers on Fidelity.com. Some stocks are not rated by all research providers. Since the model uses a number of ratings to arrive at an Equity Summary Score, only stocks that have four or more firms rating them have an Equity Summary Score. The large number of providers adds unique value across several dimensions:

- The number of IRPs yields an extensive coverage set, with over 6,000 stocks typically having at least one independent provider rating.
- With the large number of providers, Fidelity is able to provide research from firms that take very different approaches to valuation. The current research providers cover both technical and fundamental analysis, and have growth, value and momentum methodologies.
- Transparency is a core value as well, and Fidelity is always working to ensure that customers understand the research that they are using. Fidelity.com provides an overview from each research provider of its methodology and access to their stock ratings and reports. As an additional dimension of transparency, Fidelity has for many years, made available performance insight and metrics on the IRPs.

Investars is a firm that independently manages ratings databases and analyzes the performance of research providers. They have been collecting stock ratings from Fidelity's independent research providers for many years and have created a standardized ratings language (e.g., strong buy, sector outperform, neutral) across all research providers on Fidelity.com to facilitate customer understanding and comparison.

Investars collects and standardizes recommendations from IRPs that provide ratings and stock research reports on Fidelity.com and sends them at the end of each trading day to LSEG StarMine who calculates the Equity Summary Score.

Important Notes about IRP ratings and the Equity Summary Score Calculation:

- Investars receives end of trading day IRP ratings Monday through Thursday for availability by 8 am ET the next day. For IRPs that provide weekly updates after Friday market close, their rating change may not be available until Tuesday morning due to the length of weekly processing. Also if an IRP misses the deadline for daily processing, their rating change will be included in the Equity Summary Score and available on Fidelity.com the next trading day after receipt. Date of ratings is provided with the rating as well as in the research report.

Finally, we suggest caution with any performance measurement analysis, including the Equity Summary Score and related scorecards. Performance of buy/sell recommendations is only one aspect of the research offered on Fidelity.com. Although it is useful to understand a research firm's overall track record, a research firm's performance on any given stock can diverge significantly from the overall performance. There are additional factors beyond performance that any investor should consider in evaluating a research firm, such as the insights provided and the ease with which the research can be used. Performance of recommendations, while important, should not be the only factor an investor considers in evaluating research firms.

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